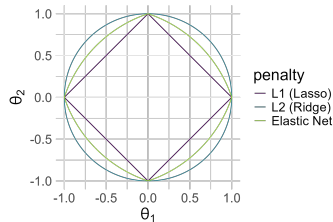
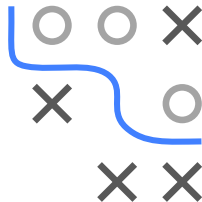


Introduction to Machine Learning

Regularization

Elastic Net and regularized GLMs

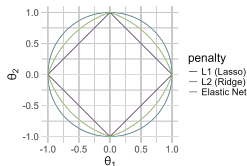


Learning goals

- Compromise between L1 and L2
- Regularized logistic regression

- Zou and Hastie 2005

A 3x3 grid with a blue path starting at the top-left corner (0,0) and ending at the bottom-right corner (2,2). The path is composed of blue line segments. Obstacles are represented by grey circles at (0,1), (0,2), and (1,2), and grey crosses at (1,0), (2,0), (2,1), and (2,2). The path starts at (0,0), goes right to (0,1), then down to (1,1), then right to (2,1), and finally down to (2,2).



- 2nd formula is simply more convenient to interpret hyperpars; λ controls how much we penalize, α sets the “L2-portion”
- Correlated features tend to be either selected or zeroed out together
- Selection of more than n features possible for $p > n$

SIMULATED EXAMPLE

5-fold CV with $n_{train} = 100$ and 20 repetitions with $n_{test} = 10000$ for setups:

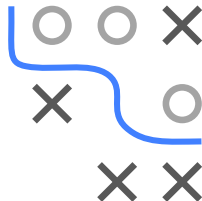
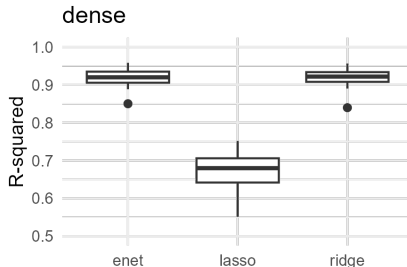
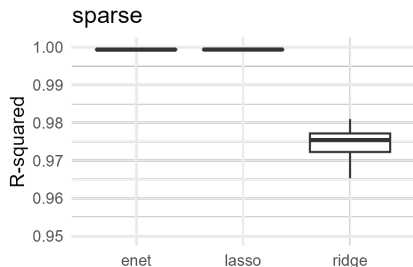
$$y = \mathbf{x}^T \boldsymbol{\theta} + \epsilon; \quad \epsilon \sim N(0, 0.1^2); \quad \mathbf{x} \sim N(0, \Sigma); \quad \Sigma_{k,l} = 0.8^{|k-l|}:$$

Lasso better for sparse features:

$$\boldsymbol{\theta} = (\underbrace{1, \dots, 1}_5, \underbrace{0, \dots, 0}_{495})$$

Ridge better for dense features:

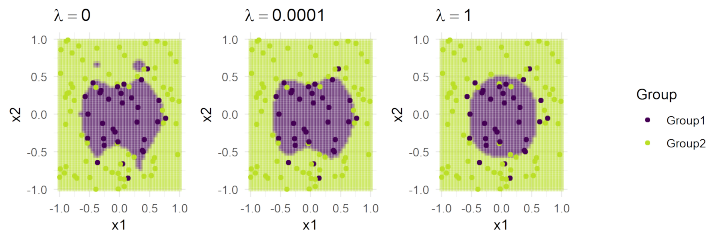
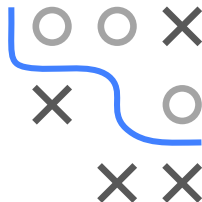
$$\boldsymbol{\theta} = (\underbrace{1, \dots, 1, 1, \dots, 1}_{500})$$



⇒ elastic net handles both cases well

REGULARIZED LOGISTIC REGRESSION

- Penalties can be added very flexibly to any model based on ERM
- E.g.: L_1 - or L_2 -penalized logistic regression for high-dim. spaces and feature selection
- Now: LR with polynomial features for x_1, x_2 up to degree 7 and L_2 penalty on 2D “circle data” below



- $\lambda = 0$: LR without penalty seems to overfit
- $\lambda = 0.0001$: We get better
- $\lambda = 1$: Fit looks pretty good